Introductory activity

Activity 1 · A pattern in powers 35 min . For an integer $n \geq 2$ and an integer a, let $r_n(a)$ be the remainder of a^{n-1} in the division by n.

1. Compute $r_5(2)$, $r_5(3)$, $r_7(2)$, $r_7(3)$ by hand, using congruences.
2. Using the Python function `pow(a, n - 1, n)`, which returns this remainder, complete the following table.

n	5	6	7	8	9	10	11
$r_n(2)$							
$r_n(3)$							

3. For which values of n do we get 1 on both rows? What do they have in common?
4. State a conjecture. Test it for $n = 13$ and $a = 5$.
5. For $n = 7$, compute $r_7(7)$ and $r_7(14)$. How should the conjecture be made more precise?

Observing is not proving: the properties conjectured here are established in the course.